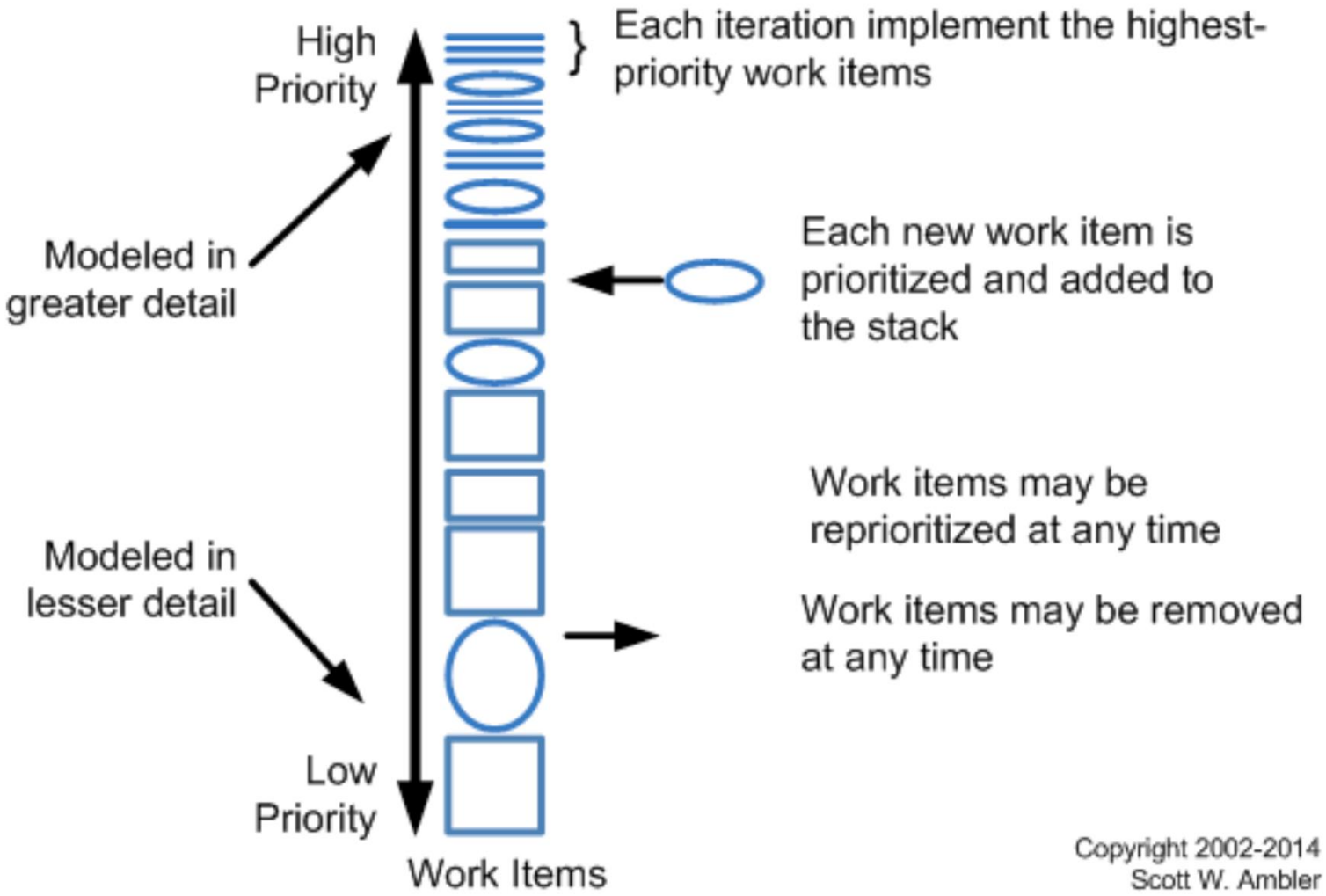
CSC 710 Software Engineering

Instructor: Dr. Tianxiao Zhang

Requirements Gathering

- The goal is to gather a list of requirements and prioritize them.
- The highest-priority requirements will be implemented first.
- These will be modeled in greater detail.
- While lower priority ones will be modeled less.



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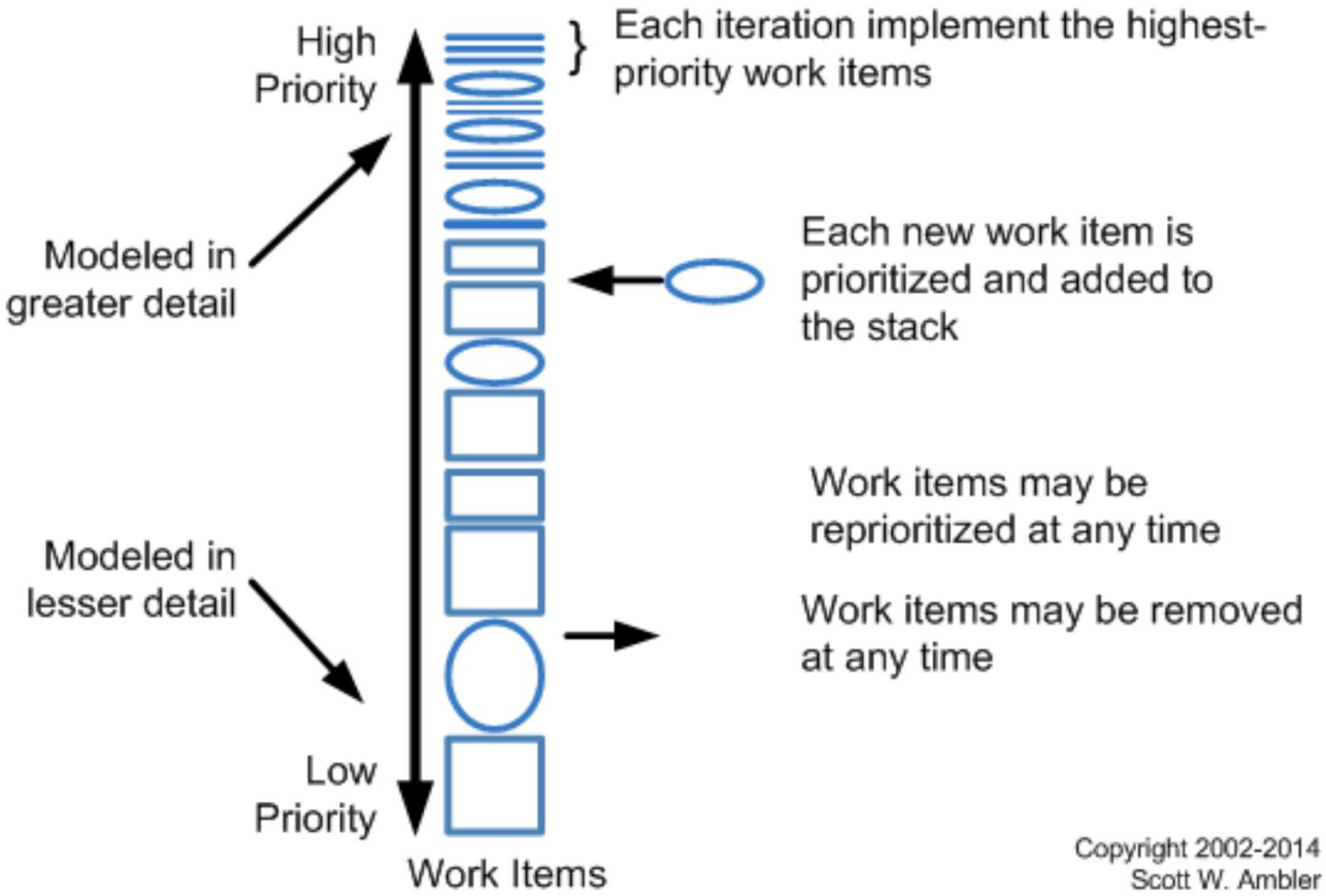
Requirements Gathering

As time goes on:
- New requirements will be added to the list and prioritized.
- Old requirements may be removed.
- The priority of requirements may be changed.



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Where Do Requirements Come From?

Your project stakeholders

- Direct or indirect users
- Managers
- Senior managers
- Operations staff members
- Support (help desk) staff members
- Testers
- Developers working on other systems that integrate or interact with yours
- Maintenance professionals

Where Do Requirements Come From?

- Project stakeholders' role is to provide requirements.
- It is the responsibility of project stakeholders to provide, clarify, specify, and prioritize requirements.
- Developer's role is to understand and implement them.
- It is the job of developers to invest the time to identify and understand those requirements.
- This concept is critical to the Agile SDLC.

Where Do Requirements Come From?

- Does that imply that you sit there in a stupor waiting for your project stakeholders to tell you what they want?
- No, of course not.
- You can ask questions to explore what they have already told you, motivating them to specify in greater detail what they want and perhaps even to rethink and modify their original requirement(s).
- You can suggest new requirements to them, the key word being SUGGEST, that they should consider and either accept (perhaps with modifications) or reject as an official requirement.

Where Do Requirements Come From?

- To identify potential requirements you may also, often with the aid of your project stakeholders, work through existing documents such as corporate policy manuals, existing legacy systems, or publicly available resources such as information on the web, books, magazine articles, or the products and services of your competitors.
- Once again, it is your project stakeholders that are the ultimate source of requirements, it is their decision and not yours.

Where Do Your Project Stakeholders Get Ideas From?

- Marketing:
  - Things that their competitors can do that they can't
  - Avoid problems that they've experienced in the past with other systems
  - Simply have a vision for a new feature
- Operations:
  - Based on the need to integrate with existing or soon-to-be existing systems
  - Motivated by an IT strategy such as reducing the number of computing platforms within your organization

Where Do Your Project Stakeholders Get Ideas From?

- The point to be made is that your project stakeholders should be formulating requirements based on a wide range of inputs.
- Something that you may want to ensure is happening by asking questions.
- But, don’t get too carried away with this (especially with senior managers or executives).
- Remember, they pay your salary!

Domain Experts

- There is significant value in involving someone with relevant expertise to the system that you are building to help identify potential requirements for the system.
- This approach is particularly valuable when your organization is building a system that includes aspects that it is not familiar with (e.g., first attempt at serving international customers via e-commerce).
- For example, in the case of an e-commerce system you might want to bring in someone with:

– international design experience

– tax law expertise

– logistics expertise

Domain Experts - Warning

- Bring external experts in for a day or two and with several project stakeholders "pick their brains" for relevant issues that you and your stakeholders may be missing due to your inexperience.
- It's a great way to make sure you're covering your bases, particularly when you are defining the initial scope for the project, ...
- ... and to get project stakeholders thinking beyond their current environment.
- However, recognize that there is a danger with this approach ...
- ... your external experts may suggest ideas that sound good but aren't actually required right now.
- In other words, you still need to work through a suggestion from an outside expert just as you would any other.

Preparation

- Remember, the source of all requirements is the project stakeholders.

- However, before seeking out the stakeholders, you need to prepare yourself.

Preparation

Reading:

- There is often a wealth of written information available to you from which you can discern potential requirements or even just to understand your stakeholders better.
- Internally:
  - Existing (albeit out of date) system documentation
  - Vision documents written to justify your project
- Externally:
  - Web sites describing similar systems
  - Competitors sites
  - Text books describing the domain in which you're currently working
- Read details in a just-in-time (JIT) manner

Preparation

Legacy Code Analysis (for existing applications):

- Work through the code, and sometimes data sources, of an existing application to determine what it does.
- Should be tempered with more interactive techniques with stakeholders.
- Strengths:
  - Identifies what has been actually implemented
- Weaknesses:
  - The actual requirements usually differ from what you currently have
  - It can be difficult to extract requirements from legacy code, even with good tools

Active Stakeholder Participation

- Once you are prepared (and don’t spend too much time preparing) …
- … actively involve stakeholders in modeling of requirements.
- Strengths:
  - Highly collaborative technique
  - The people with the domain knowledge define the requirements
  - Information is provided to the team in a timely manner
  - Decisions are made in a timely manner
- Weaknesses:
  - Many stakeholders need to learn modeling skills.
  - Stakeholders often aren't available 100% of the time.

Active Stakeholder Participation

- Adopt inclusive models (e.g., white board – get stakeholders involved in process)
- Explain the techniques
- Adopt stakeholder terminology
- Turn stakeholders into developers
- Keep it fun
- Obtain management support
- Gather details just-in-time (JIT)
- Smaller is better
- Treat requirements like a prioritized stack

Techniques for Eliciting Requirements

- Here are a few techniques for eliciting requirements from stakeholders:
  - On-Site Customer
  - Face-to-Face Interviews
  - Electronic Interviews
  - Focus Groups
  - Joint Application Design (JAD)
  - Observation
- Each technique has trade-offs.
- You'll need to learn several if you want to become adept at eliciting requirements.

On-Site Customer

- In Agile XP, the customer role is filled by one or more people who are readily available to provide domain-related information to the team and to make requirements-related decisions in a timely manner.
- Other Agile methods improve on this by getting more stakeholders involved
- Strengths:
  - Collaborative technique
  - Information is provided to the team in a timely manner
  - Decisions are made in a timely manner
- Weaknesses
  - Stakeholders need to be educated in their role
  - Stakeholders need to be available on-demand

Face-to-Face Interviews

- Meet with someone to discuss their requirements.
- Although interviews are sometimes impromptu events, it is more common to schedule a specific time and place to meet and to provide at least an informal agenda to the interviewee.
- It is also common to provide a copy of your interview notes to the interviewee, along with some follow up questions, for their review afterward.
- One danger of interviews is that you'll be told how the person ideally wants to work, not how they actually work.
- You should temper interviews with actual observation.
- Be prepared to follow-up
- Hold the interview at a whiteboard, so that you can sketch as you talk, turning the interview into a model storming session
- Actively listen to what they're saying

Face-to-Face Interviews

- Strengths:
  - Collaborative technique
  - You can elicit a lot of information quickly from a single person
  - People will tell you things privately that they wouldn't publicly
- Weaknesses:
  - Interviewing skills are difficult to learn

Electronic Interviews

- Interview a person over the phone, through video conferencing, or via email.
- Face-to-face interviews should be preferred over electronic ones
- Strengths:
  - Supports environments with dispersed stakeholders
  - Provides a permanent record of the conversation
- Weaknesses:
  - Restricted interaction technique
  - Limited information can be conveyed electronically
  - Risky when it is your only means of communication

Focus Groups

- You invite a group of actual and/or potential end users, if one exists, to brainstorm requirements for the new one.
- Hold it in a room with whiteboards or flip chart paper so people can draw as they talk.
- Strengths:
  - Collaborative technique
  - Significant amounts of information can be gathered quickly
  - Works well with dispersed stakeholders
  - Works well when actual users do not yet exist
- Weaknesses:
  - Lots of unimportant information will be conveyed
  - It's difficult to identify the right people
  - Focus groups can be diverted by a single strong-willed individual

Joint Application Design (JAD)

- A JAD is a facilitated and highly structured meeting that has specific roles of facilitator, participant, scribe, and observer.
- JADs have defined rules of behavior including when to speak, and typically use a U-shaped table.
- It is common practice to distribute a well-defined agenda and an information package which everyone is expected to read before a JAD.
- Official meeting minutes are written and distributed after a JAD, including a list of action items assigned during the JAD that the facilitator is responsible for ensuring are actually performed.
- Improvements:
- – Loosen the rules about when people can talk
- – Hold it in a room with whiteboards or flip chart paper so people can draw as they talk

Joint Application Design (JAD)

- Strengths:
  - Facilitator can keep the group focused
  - Significant amounts of information can be gathered quickly
  - Works well with dispersed stakeholders
- Weaknesses:
  - Restricted interaction technique
  - Facilitation requires great skill

Observation

- Sit and watch end users do their daily work to see what actually happens in practice.
- Usually differs from the idealistic view which they tell you in interviews or JADs.
- Observation is best done passively.
- Take notes and then ask questions after an observation session to discover why the end users were doing what they were doing at the time.

Observation

- Strengths:
  - Helps to identify what people actually do
  - Provides significant insight to developers regarding their stakeholder environments
- Weaknesses:
  - Restricted interaction technique
  - It is hard to merely observe, you also want to interact
  - Seems like a waste of time because you're "just sitting there"
  - Can be difficult to get permission

Requirements Gathering Challenges

- Limited access or no access to project stakeholders
- – Use one of the techniques for eliciting requirements that doesn’t require the software engineer and stakeholders to physically meet in the same location (i.e., colocation).
- Geographically dispersed project stakeholders
- – Use one of the techniques for eliciting requirements that doesn’t require colocation.
- Project stakeholders only have a vague idea of what they want
- – Start small.
- Project stakeholders change their minds
- – People change their minds.
- Conflicting project stakeholder priorities
- – Negotiate

Requirements Gathering Challenges

- Too many project stakeholders want to participate
  - It happens, especially if your project is a “political winner”
  - Try to keep only the ones you know will be helpful
- Project stakeholders prescribe technology solutions
  - Ask the stakeholder “if you had that technology in place, what would it do for you that's important” or “how would you use this technology” in an effort to identify the actual requirements
- Project stakeholders are unable to see beyond the current situation
  - People have been doing their jobs the same way using the same tools for years.
  - They may not have ever seen another way to do things nor have they thought it could be different.
  - Talk about the current situation with them, to identify what works well and what doesn't, and to explore how the current situation came about.
  - Then both of you can see how it might be improved.

Requirements Gathering Challenges

- Project stakeholders are afraid to be pinned down
  - They don’t want you coming back later saying, “But you signed off on this …”
  - Impress on your project stakeholders that you are ready to embrace change, that you are taking an iterative and incremental approach that reduces the cost of change
- Project stakeholders don't understand modeling artifacts
  - Your first step is to identify artifacts that your project stakeholders will understand.
  - This is where the practice Use The Simplest Tools helps.
- Developers don't understand the problem domain
  - Developers need to invest the time to learn the domain.
  - The stakeholders don’t understand your environment either.
  - This is where the participatory interaction between developers and stakeholders on a day-to-day basis helps

Requirements Gathering Challenges

- Project stakeholders are overly focused on one type of requirement
  - Sometimes you will find that your project stakeholders are providing you with great usage requirements, but are falling short when it comes to non-functional requirements (or vice versa).
  - Add new stakeholders that can help you with the missing type of requirement.
- If all else fails:
  - Ask your team lead or project leader to get involved.
  - If you are the team lead or project leader -- Call in the person funding the project.

What Is Brainstorming?

According to the Merriam-Webster dictionary:

- A group problem-solving technique that involves the spontaneous contribution of ideas from all members of the group.
- The mulling over of ideas by one or more individuals in an attempt to devise or find a solution to a problem.

What Does It Have To Do With Software Engineering?

To generate requirements

As part of gathering requirements for:

- On-Site Customer
- Face-to-Face Interviews
- Electronic Interviews
- Focus Groups
- Joint Application Design (JAD)

Question Brainstorming or Starbursting

- Team spends time generating as many requirements as possible.
- Without trying to come up how to implement the requirements.
- Involves asking questions using the 5 W's and 1 H:
  - Who (e.g., who is the system for)
  - What (e.g., what is the system for, what problem does it solve)
  - Where (e.g., where will the system be deployed, laptop, phone, etc.)
  - When (e.g., when will the system be used, in school, during evenings)
  - Why (e.g., why is the system necessary, why would someone buy this)
  - How (e.g., how will the system be used)
- Answers to questions generate requirements.
- Ensures that the problem has been thought of from multiple angles

Hybrid Brainstorming & Brain-Netting

- Combine individual and group brainstorming efforts
- Leveraging the benefits of both asynchronous and synchronous idea generation
- Gather diverse range of ideas from participants working independently, and then together for collaborative refinement and prioritization

Hybrid brainstorming

- Start with individual brainstorming session
- Followed by a collaborative group session to share, discuss, and refine ideas

Brain-netting (works well with remote teams)

- Online brainstorming session done asynchronously
- Participants contribute ideas on a shared platform (e.g., Discord) over a set period

Mind Map & Silent Circuit Brainstorming

Mind map

- Create a visual representation of ideas branching out from a central topic
- Helps participants see connections and relationships between different variables

Silent circuit

- Quiet brainstorming method
- Participants add ideas to a canvas or board in silence
- Ensures inclusivity quieter team members to contribute equally
- Minimizes the influence of dominant voices

Crazy 8s & 6-3-5 Brainwriting

- Time-boxed brainstorming techniques
- Aim for coming up with ideas in a rapid succession
- Encourages participants to think creatively under time constraints

Crazy 8s

- Divide a sheet of paper into eight sections
- Each participant spends one minute on each section to sketch or detail an idea
- Results in eight ideas in eight minutes

6-3-5 brainwriting

- 6 participants write down 3 ideas in 5 minutes
- Repeated for 6 rounds
- Results 108 ideas – 6 participants X 3 ideas X 6 rounds
- Easily modified for more or less participants

Reverse & Round Robin Brainstorming

- Looking at problems from different perspectives
- Encourage critical thinking
- Building on each other's ideas - collaborative improvement

Reverse brainstorming

- Participants think about how to make a problem worse
- Uncover new insights that can be flipped into positive solutions

Round robin brainstorming

- Divide participants into small groups
- Each person passes their ideas to the next member of the team for further suggestions or counterpoints

Individual Brainstorming Techniques

To prepare for group brainstorming sessions

- CSC 710 team meetings
- On-Site Customer
- Face-to-Face Interviews
- Electronic Interviews
- Focus Groups
- Joint Application Design (JAD)

Individual Brainstorming Techniques

As part of group brainstorming techniques

- Question Brainstorming or Starbursting
- Hybrid Brainstorming & Brain-Netting
- Mind Map & Silent Circuit Brainstorming
- Crazy 8s & 6-3-5 Brainwriting
- Reverse & Round Robin Brainstorming

Individual Brainstorming Techniques

Remote work brainstorming sessions

- More asynchronous
- More individual

Benefits of Individual Brainstorming

- Listening to other team members can block you from generating your own ideas.
- Not all team members strictly follow the rules of brainstorming (i.e., judging or criticizing others) discouraging creativity
- In solo brainstorming, these issues are completely removed
- You can consider controversial or seemingly imperfect ideas
- Explore them at your own pace.

Zero Draft Brainstorming

- Popular one among writers for generating innovative ideas
- 1. Write down any initial thoughts, ideas, emotions, or questions about the product.
- 2. Organize thoughts in a table or idea management software to allow you to see a bigger picture.
- 3. Use SCAMPER to improve ideas (e.g., requirements for product).

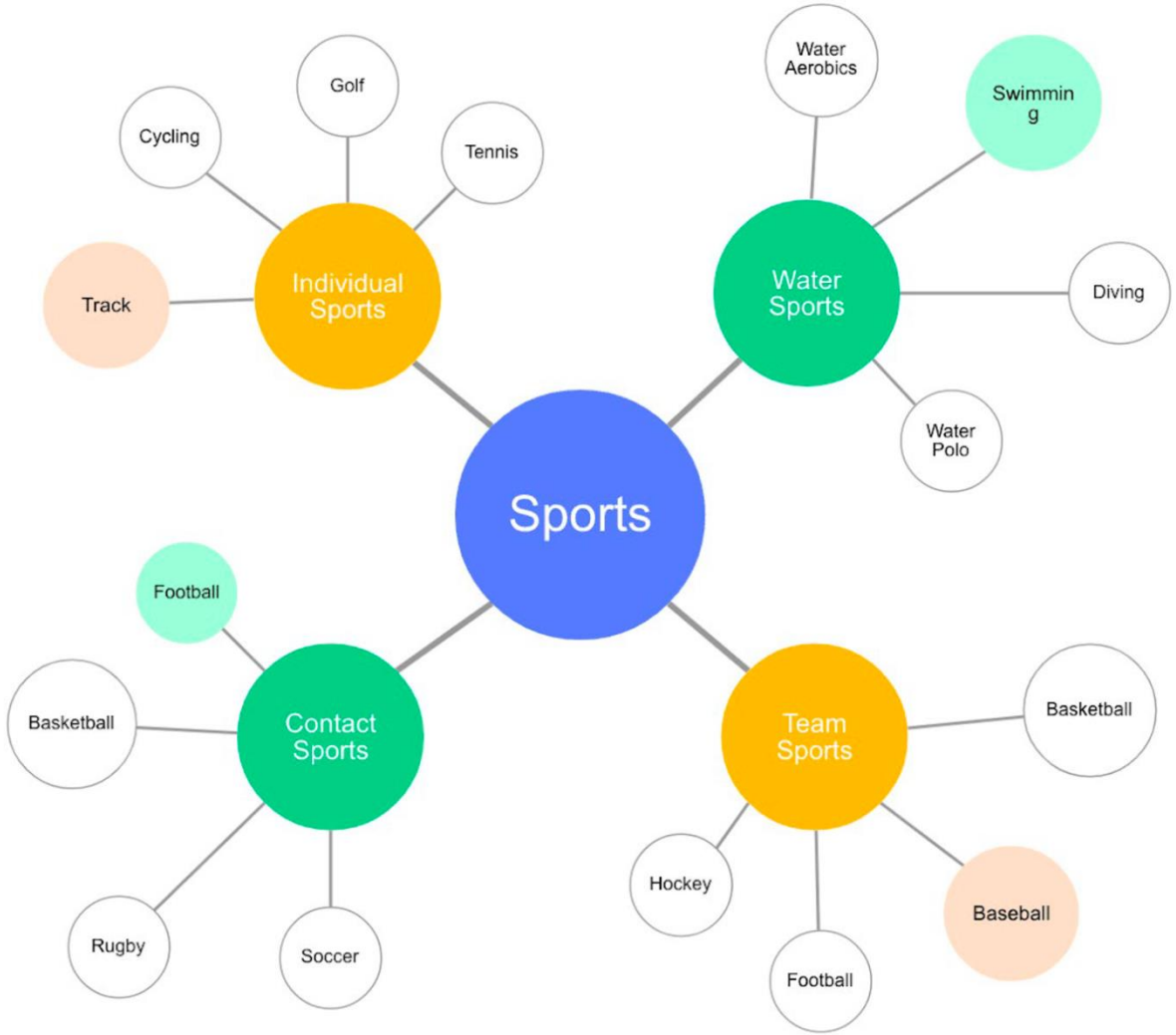
Zero Draft Brainstorming

SCAMPER

- **S**ubstitute - what requirements can be replaced
- **C**ombine - merge requirements into a new requirement for the product
- **A**dapt - what steps are necessary to change or improve the product
- **M**odify - how can the appearance and value of the product be altered
- **P**ut to another use - can the product be used in a unique way not yet thought of
- **E**liminate - what might make the product too complex
- **R**everse - reorganize the product's requirements to see it from a new light

Mind Mapping

- Visually organize data and information
- Popular for those who interpret data better visually than in a list
- Organized around a central idea, it works like the branches of a tree, where ideas and sub-ideas associated with the main idea branch off from it.



Question